Thus far, we've focused on the inferential task of point estimation: providing a single (perhaps "best") guess for an unknown parameter in a statistical model. But in doing so, we essentially ignore the fact that there is randomness in the data, and thus variability in our estimator. It's true that we calculate the variance of our estimators when obtaining MSE or comparing different unbiased estimators against each other. But do we ever present the notion of variability when we claim that $\hat{\theta}$ is our estimator for θ? No! We might want to express how confident we are that $\hat{\theta}$ is close to the true θ.

We know our estimators are random variables because they are functions of the observable random variables that we call the data X_i. As a result, the variability of our estimators are inherited from the data. We have a special name for the distribution of a statistic: the **sampling distribution** (defined shortly). Why does it get a special name? In order to clearly differentiate between the distribution of a sample (i.e. the joint density in the statistical model) and the distribution of the statistic derived from that sample (which is why it's called the 'sampling' distribution).

Let $X_1, \ldots, X_n$ be an IID sample from distribution parameterized by unknown θ or $\boldsymbol{\theta}$. Let $T(\vec{X})$ be a statistic. The distribution of T for a given value of θ is called the **sampling distribution** of T.

- e.g.: If $X_1, \ldots, X_n|\mu, \sigma^2 \overset{\text{iid}}{\sim} N(\mu, \sigma^2)$, then $\bar{X}|\mu, \sigma^2 \sim N\left(\mu, \frac{\sigma^2}{n}\right)$. (*You may have shown this in MATH/STAT 0310*). The sampling distribution of $\bar{X}$ is useful because we can use it to answer questions such as: What is the probability that $\bar{X}$ is within 1 standard deviation of its mean? That is, I can calculate $P\left(|\bar{X} - \mu| \leq \frac{\sigma}{\sqrt{n}}\right)$. If I'm trying to estimate μ using $\bar{X}$, this might a very important/useful quantity to obtain!

The utility of using sampling distributions comes from their ability to tell use *how much* we can learn about θ given our data. This will help us decide between different statistics to use as estimators. Additionally, once we choose our statistical model and a statistic, we in theory have access to the sampling distribution of that statistic. Thus, we can assess its theoretical performance before ever seeing real data. That is so useful from a resources perspective!

Life would be nice if all sampling distributions were Normal, but that would be too easy! Suppose I want the sampling distribution of the sample variance S^2. The Normal distribution is obviously an inappropriate candidate for the sampling distribution of S^2 because we know that that statistic cannot be negative! So, we may have to obtain sampling distributions on a case-by-case basis.

But don't be scared! Sometimes, sampling distributions of statistics are easy to find/derive, especially when our data are Normally distributed (which is one of the most common scenarios in real life). This week, we will focus on learning new named distributions that are related to the Normal distribution. Why? These distributions commonly arise as sampling distributions for some statistics we will encounter.